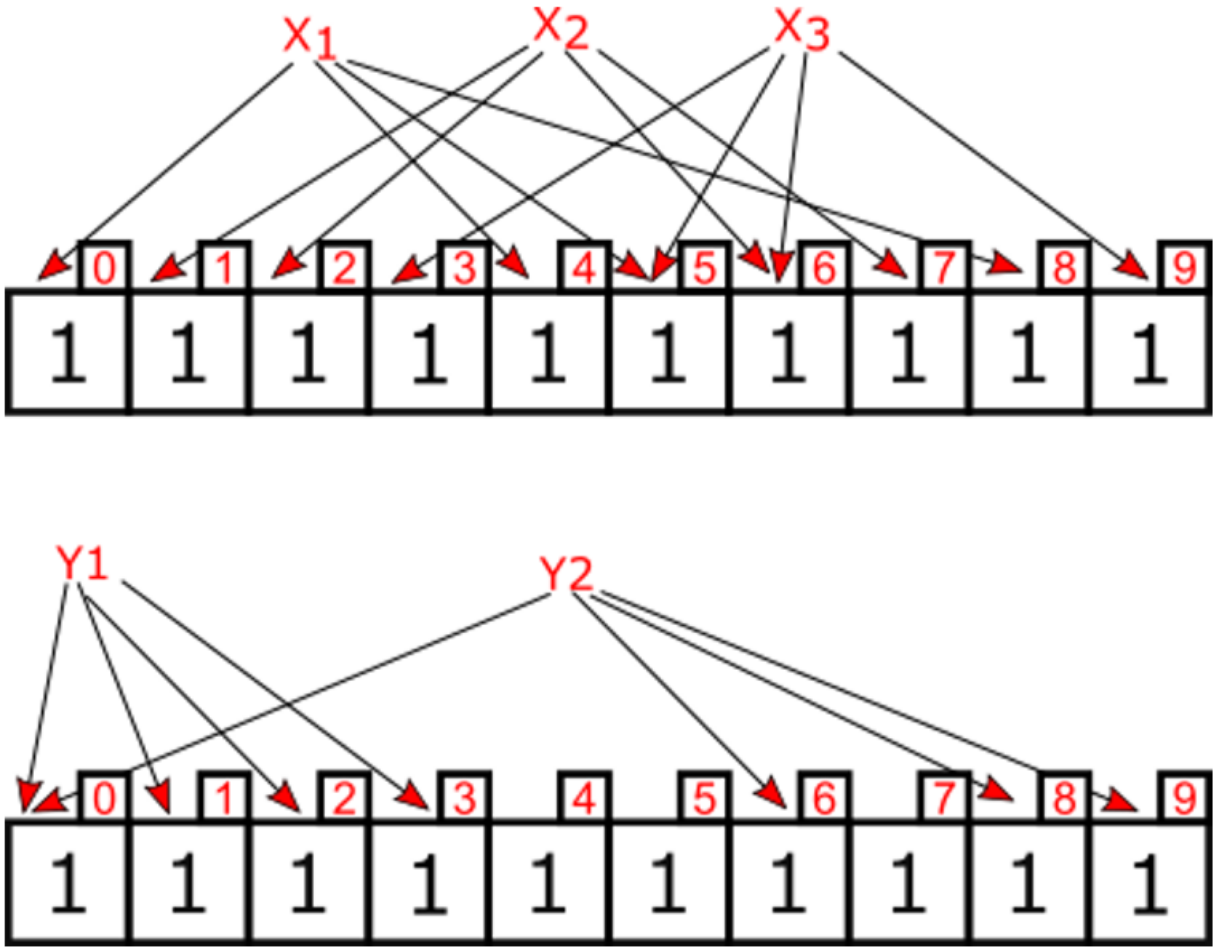


Figure 4.6 Bloom filter saturation. In the example, `m=10`, so the bloom filter has 10 bits, and `k=4`, so each element is stored with 4 bits. We can saturate the filter by adding 3 elements, setting all bits to `0`. Then even if we look up Y_1 and Y_2, which were not added to the filter, still we get false positive. Although the example is an extremization (with `k` unrealistically too large for the value of `m`), it illustrates the mechanism that leads to saturation, or anyway to degraded precision.

As already mentioned, Bloom filters are a tradeoff between memory and accuracy. If you are going to create an instance